

SMART INDIA HACKATHON 2026

Deep Learning-Based Super-Resolution Mapping (SRM) from Medium-Resolution Satellite Imageries

- Problem Statement ID – SIH 26142
- Theme- Space Technology
- PS Category- Software
- Team ID - 178177
- Team Name- PlotSphere



REPORT



GITHUB



PROPOSED SOLUTION



4× Super-Resolution: Takes four 10 m Sentinel-2 bands (B02, B03, B04, B08) and outputs them on a **2.5 m grid** (4x finer).



Model: EDSR (residual CNN) with **1.52 M parameters**. Learns only the missing fine detail on top of a bicubic copy, keeping colours and brightness consistent with the satellite image.



Geospatial Consistency: Output keeps the same coordinate system and location as the input. Measured shift is **0.002 px**.

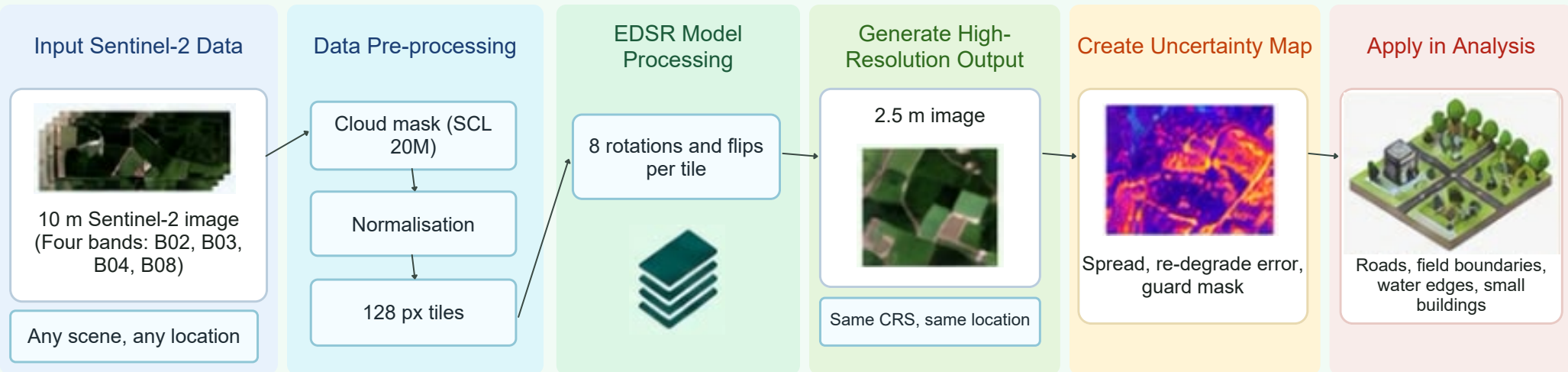


Uncertainty Output: Every product comes with a 3-layer uncertainty file marking where detail is guessed and not observed.



ENHANCED REMOTE-SENSING IMAGE PROCESSING

Sentinel-2 Super-Resolution Workflow



INNOVATION EDGE



Two-Stage Training: 2,265 Indian patches, then jointly fine-tuned with real Sentinel-2/2.5 m reference pairs and Indian data.



Physics-Aware Losses: Combines pixel, structural, spectral, edge and re-degradation losses to improve spatial detail while preserving **spectral** info.



Validated Uncertainty: Uses 8× test-time augmentation to estimate uncertainty; top **10%** most uncertain pixels show **≈ 4× higher error**.



Reliability Guard: Bright or unreliable pixels fall back to bicubic output to prevent false colours.



Real-World Validation: Tested on synthetic data and real 2.5 m reference imagery.



REAL-WORLD SUPER-RESOLUTION RESULT

Ahmedabad / Scene 2_2025-04-03 / patch 9



Roads, field boundaries and the edge of the settlement come back clearly.



PSNR goes from **25.08 dB** → **27.21 dB**



REAL 2.5 m REFERENCE RESULT (NEVER SEEN IN TRAINING)

SPOT 2.5 m reference



Real SPOT 2.5 m reference that the model never saw in training. The uncertainty (last panel) sits on the drainage lines.



Detail correlation goes from **0.391** → **0.469**

PSNR

20.92 → **21.50 dB**



WHY IT MATTERS



Sharper Land Insights

Clearer roads, field boundaries and settlements.



Better Environmental Monitoring

Improved tracking of urban growth, agriculture and water bodies.



Actionable for India

High-resolution, reliable maps for planning, disaster response and sustainable development.



1.52M
Parameters

EDSR model



95.5 s

10 km × 10 km scene
(RTX 3050, 4 GB)



0.002 px

Measured shift
(same location)



Top 10%
≈ **4× error**

Uncertainty
(8× TTA)



4× finer

10 m → 2.5 m
grid resolution

Free Sentinel-2 data → leak-free Indian training set → two-stage EDSR → 2.5 m GeoTIFF with an uncertainty map

DATA ACQUISITION & STUDY AREAS

- Source:** Copernicus Data Space Ecosystem, Sentinel-2 Level-2A
- Bands:** B02, B03, B04, B08 at 10 m and SCL 20 m
- Study areas:** Ahmedabad, New Delhi, Nashik, Guwahati (four different climates)
- 5 dates per city**, January 2025 to June 2026 — 20 scenes in total
- 2,265 patches** of 512 × 512 px (5.12 km each side)
- Real 2.5 m references:** opensr-test benchmark, 119 Sentinel-2 / 2.5 m pairs (NAIP, SPOT/WorldStrat, Spain)

PRE-PROCESSING

- Cloud Masking:** Opaque clouds stored as no-data and left out of every loss and metric. Patches with less than 90% clear pixels are dropped.
- Normalisation:** Per-band p1/p99 scaling shared across all cities
- Training pairs:** Each 10 m patch is blurred and shrunk 4× (Wald protocol) to make the low-res input, since no Indian 2.5 m imagery is free
- Leak-free split:** Whole cities held out for testing (Nashik, Guwahati), plus the last 15% strip of each city for validation. Zero overlap checked.

PATCHES PER CITY

2,265 patches of 512 × 512 px, 5 dates per city

City	Patches	Type
Ahmedabad	539	Training cities
Guwahati	192	Held out for testing
Nashik	307	Held out for testing
New Delhi	1,227	Training cities

54% of the data comes from New Delhi, so Nashik was held out for testing instead of Delhi.

MODEL DEVELOPMENT & VALIDATION

STAGE 1 EDSR trained on Indian patches · 60 epochs

STAGE 2 Fine-tuned 30 epochs · each batch = 4 real 2.5 m + 4 Indian

- Loss =** 1.0 Charbonnier + 0.1 SSIM + 0.05 spectral angle + 0.05 edge + 0.1 regrade
- Tested on **unseen cities** and on **29 real 2.5 m images** never used in training
- Metrics:** PSNR, SSIM, spectral angle (SAM), ERGAS, NDVI error, real detail correlation, with 95% bootstrap confidence intervals
- Uncertainty checked against the real error** (Spearman correlation)

Training loss weights

Loss Component	Weight
Charbonnier	1
SSIM	0.1
Spectral	0.05
Edge	0.05
Re-degrade	0.1

TECHNOLOGY STACK

Acquire
Copernicus S-2

Process
GDAL · rasterio

Train
PyTorch EDSR

Deliver
GeoTIFF · QGIS

10 m

4×

2.5 m

2,265 patches

29 unseen tests

4 GB laptop GPU

END-TO-END FRAMEWORK Training → scene inference

A. DATA & TRAINING

B. INFERENCE (any Sentinel-2 scene)

EDSR generator: 1.519 M parameters, no batch normalisation

bicubic upsampling of the input — the network learns only the missing high-frequency residual

Top: data preparation and two-stage training. Bottom: scene-level inference with uncertainty, and the EDSR model inside it.

Same free satellite, same 5-day revisit — 16× more pixels over the same ground, and an honest map of what was inferred.



FROM 10 m INPUT TO DECISION-READY MAP

01

10 m SENTINEL-2 INPUT

- Free, every 5 days, covers all of India
- But at 10 m, one pixel covers a **two-lane road, a field bund or a small building**
- Small buildings, narrow roads, field boundaries and water edges get lost

02

AI SUPER-RESOLUTION

- EDSR model adds fine detail on a **2.5 m grid**
- Colours and band ratios kept: spectral-angle error **19% lower**, NDVI error **22% lower** than bicubic
- No location shift (**0.002 of a pixel**)

03

2.5 m OUTPUT + UNCERTAINTY

- **+1.48 dB PSNR** over bicubic on held-out Indian data (95% CI 1.44–1.52), better on **100% of 238 patches**
- Real sub-10 m detail on unseen 2.5 m references: **+0.030** (CI 0.013–0.046), 79% of images, 100% on WorldStrat SPOT
- Every pixel has an uncertainty value; the most uncertain 10% carry about **4× the error**

04

READY FOR ANALYSIS

- **GeoTIFF in the same CRS**, opens directly in QGIS or ArcGIS
- 10 km × 10 km scene in **95.5 s on a laptop GPU**, so district offices can run it without a server



+1.48 dB

PSNR over bicubic
(held-out India)



100%

of 238 patches
improved



22%

lower NDVI error
than bicubic



95.5 s

per 10 km scene
(laptop GPU)



REAL-WORLD IMPACT

Real Nashik scenes, never used in training



AGRICULTURE

19–22% lower NDVI error

- Sharper field boundaries for parcel-level crop monitoring
- NDVI error **19–22% lower** than bicubic on held-out Indian data
- Field boundaries • Crop monitoring • Crop stress

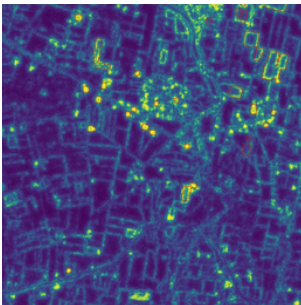
Sentinel-2 input (10 m)



Our output (2.5 m grid)



Uncertainty + guard (red)



Nashik farmland, 4 June 2026: parcels, polyhouses, roads and the river channel sharpen. The guard (red, 2.6% of the scene) fires on bright polyhouse roofs.



URBAN MAPPING

Honest flags, never invents

- Roads, railways and open spaces come out clearly
- Dense old-city blocks are **not invented**; they are flagged as uncertain
- Roads • Infrastructure • Urban growth

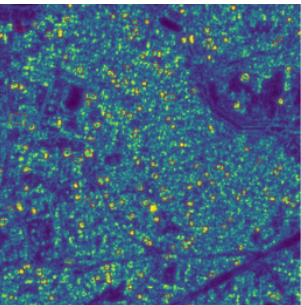
Sentinel-2 input (10 m)



Our output (2.5 m grid)



Uncertainty + guard (red)



Nashik city: roads, the railway and open spaces sharpen. Dense old-city blocks stay smooth and are marked uncertain.



WATER & DISASTER RESPONSE

Trust map per pixel

- River and reservoir edges are sharper
- Uncertainty and guard layers show which areas should not be trusted for a decision
- Flood extent • Damage mapping • Affected areas



ENVIRONMENT

Every 5 days revisit

- Better land-cover boundaries for change detection over time
- Water boundaries • Land cover • Change detection

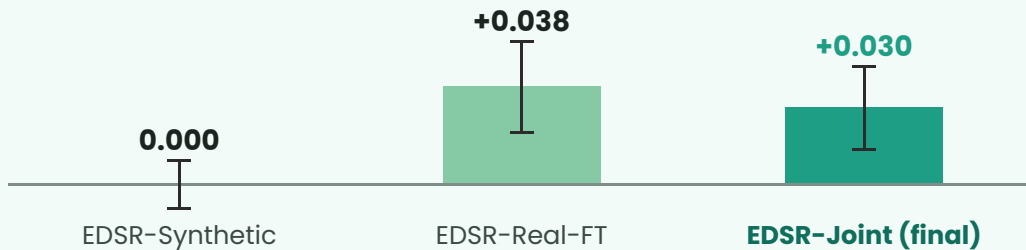
Only the joint model beats bicubic on both tests

Gain over bicubic, with 95% confidence intervals

Indian test, 238 held-out patches • PSNR gain (dB)



Real 2.5 m references, 29 unseen • detail-correlation gain



Field-boundary F1 and water-mask IoU are the next things we will measure; we do not claim them yet.

WHO CAN USE IT

State agriculture departments for parcel-level crop checks, **urban local bodies** for road and growth mapping, and **disaster management authorities** for flood edges — without buying high-resolution imagery.

Free data, an open-source stack and one laptop GPU — and every risk we found, measured and handled.



IMPLEMENTATION FEASIBILITY

DATA

- Sentinel-2 is **free** and covers India every 5 days
- Real 2.5 m pairs are openly available (opensr-test, WorldStrat), so we **did not need to buy imagery**

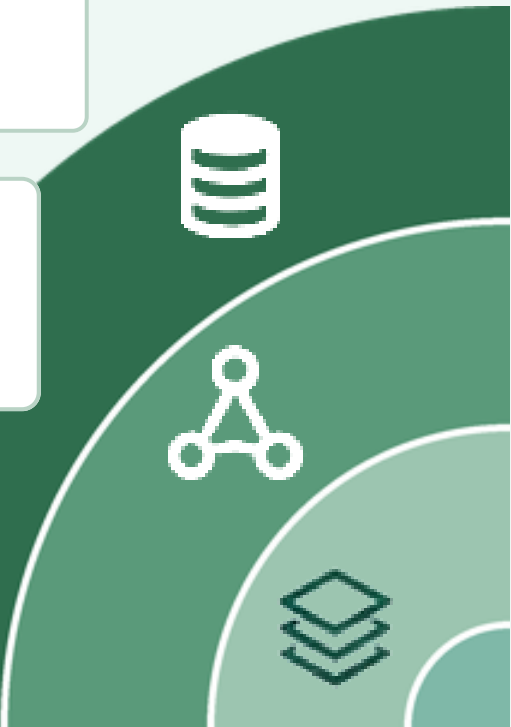
TECHNOLOGY

- **All open-source:** PyTorch, rasterio, GDAL, QGIS
- **Small model:** 1.52 M parameters, 6.1 MB file
- Stage-1 training takes **under 1.2 hours** on a laptop RTX 3050

DEPLOYMENT

- **Tile-based:** 128 px tiles with a 16 px overlap, so any scene size fits in 4 GB of GPU memory
- 10 km × 10 km scene in **95.5 s**
- Output is a standard GeoTIFF plus an uncertainty GeoTIFF — **nothing extra to install**

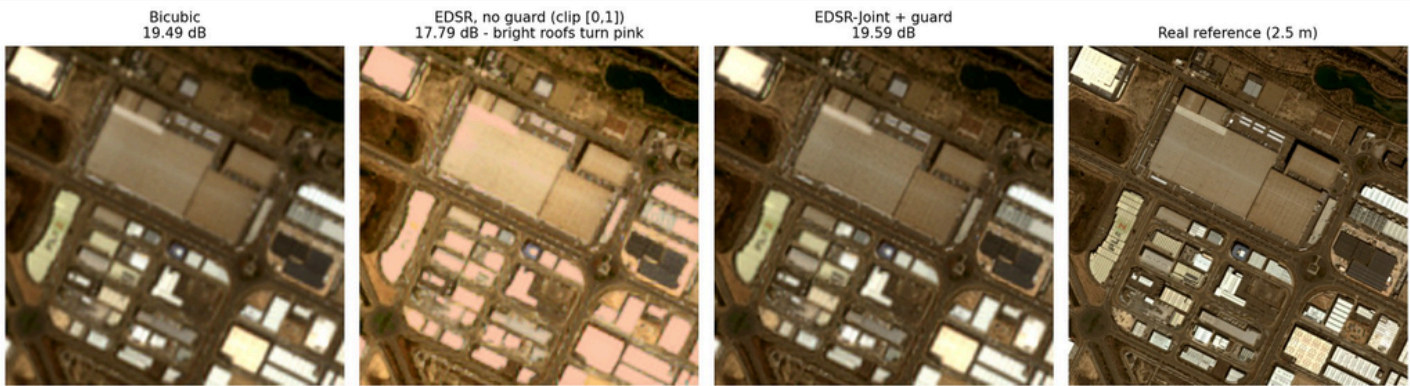
₹0 data cost • fully open-source stack • one laptop GPU • every number in our report traces back to a saved result file



MAJOR HEADWINDS & RISKS

Challenge	What we did about it	Status
No free 2.5 m imagery over India	Trained on downscaled Indian data first , then fine-tuned on 90 real pairs from the USA and Spain	Handled
Clouds and haze	Opaque clouds masked out of training and testing. Thin haze is not masked yet; next step: SCL cirrus and aerosol screening	Handled
Model inventing detail	Tested a GAN: it drew fake parcel outlines and lost accuracy, so we dropped it. The uncertainty map flags guessed detail	Handled
Bright roofs turning pink	Out-of-range guard: about 2% of pixels fall back to bicubic and are flagged. That image went from 17.79 to 19.59 dB	Fixed
Weaker results in humid, hilly areas	Guwahati gain is +0.72 dB vs +1.25 dB in Nashik. Plan: add humid, coastal and hilly cities	Planned
Compute cost	Small CNN instead of a diffusion model; runs on a laptop GPU	Handled

Our own failure, found and fixed: bright roofs turning pink

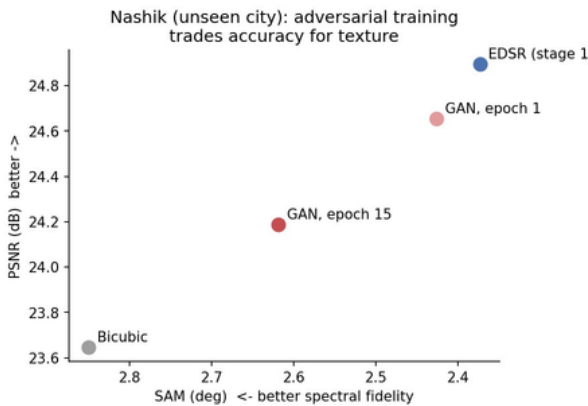


Spain urban test image. Without the guard, bright roofs turn pink and lose 1.7 dB. With the guard the model beats bicubic (19.59 vs 19.49 dB).

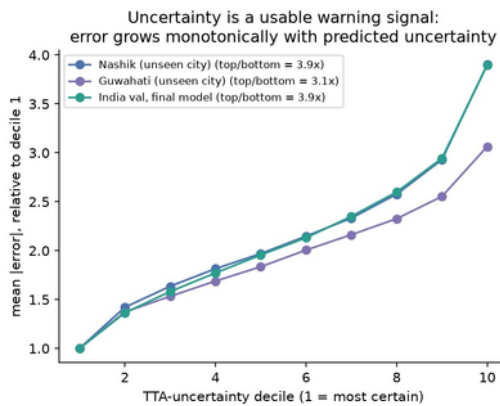
WHAT WE CLAIM AND WHAT WE DON'T

- **The output grid is 2.5 m.** Real detail goes up, but pixel accuracy on real 2.5 m data is about the same as bicubic. It is **not the same as native 2.5 m imagery**.
- **No Indian 2.5 m reference yet.** Next: validate with Cartosat or aerial imagery.

2.5 m output grid	Yes
More real detail	Yes
Pixel accuracy vs bicubic	Better
Same as native 2.5 m imagery	No



The longer the GAN trains, the worse both PSNR and spectral accuracy get. EDSR stays on top.



Error rises steadily with uncertainty in every setting, so the uncertainty map can be trusted as a warning.

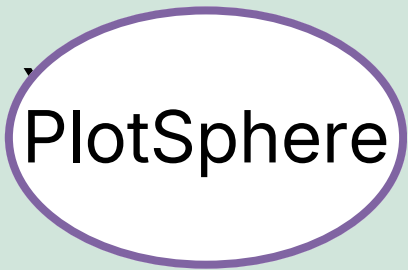
VISIBILITY

1 Existing 10 m Sentinel-2 imagery

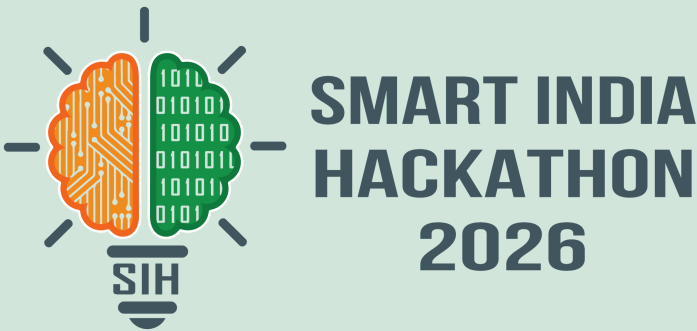
2 Automated EDSR pipeline (95.5 s per scene)

3 2.5 m GeoTIFF + uncertainty map

4 Agriculture, urban planning, disaster response



RESEARCH AND REFERENCES



1 MODEL & TRAINING

what we built on

- 1

Lim et al., 2017
Enhanced Deep Residual Networks for Single Image Super-Resolution (EDSR)
CVPR Workshops

Our backbone
- 13

Shi et al., 2016
Real-Time Single Image and Video SR Using an Efficient Sub-Pixel CNN
CVPR

4× upsampler
- 16

Charbonnier et al., 1994
Two Deterministic Half-Quadratic Regularization Algorithms for Computed Imaging
IEEE ICIP

Pixel loss

2 DATA & VALIDATION

evidence sources

- 5

Drusch et al., 2012
Sentinel-2: ESA's Optical High-Resolution Mission for GMES
Remote Sensing of Environment

Input sensor
- **Copernicus Data Space Ecosystem**
Sentinel-2 Level-2A surface reflectance products
ESA / European Commission

Indian data
- 3

Aybar et al., 2024
A Comprehensive Benchmark for Optical Remote Sensing Image SR (opensr-test)
IEEE Geoscience and Remote Sensing Letters

Real 2.5 m test
- 4

Cornebise et al., 2022
Open High-Resolution Satellite Imagery: The WorldStrat Dataset
NeurIPS Datasets & Benchmarks

SPOT references

3 EVALUATION METRICS

how we measured

- 2

Wald et al., 1997
Fusion of Satellite Images of Different Spatial Resolutions: Assessing the Quality
Photogramm. Eng. & Remote Sensing

Wald protocol
- 6

Wang et al., 2004
Image Quality Assessment: From Error Visibility to Structural Similarity
IEEE Trans. Image Processing

SSIM
- 10

Kruse et al., 1993
The Spectral Image Processing System (SIPS)
Remote Sensing of Environment

Spectral angle
- 11

Wald, 2000
Quality of High Resolution Synthesised Images: Is There a Simple Criterion?
Proc. Fusion of Earth Data

ERGAS

4 GENERATIVE (GAN) VARIANT: TESTED AND REJECTED

ablation, not the final model

- 7

Wang et al., 2018
ESRGAN: Enhanced Super-Resolution GANs
ECCV Workshops

GAN design
- 14

Miyato et al., 2018
Spectral Normalization for GANs
ICLR

Discriminator
- 9

Blau & Michaeli, 2018
The Perception-Distortion Tradeoff
CVPR

Why we rejected it
- 8

Jolicœur-Martineau, 2019
The Relativistic Discriminator
ICLR

GAN loss
- 15

Isola et al., 2017
Image-to-Image Translation with Conditional Adversarial Networks
CVPR

PatchGAN

5 TERMINOLOGY

problem framing

- 12

Atkinson, 1997
Mapping Sub-Pixel Boundaries from Remotely Sensed Images
Innovations in GIS 4, Taylor & Francis

"SRM" term
- Classic SRM means sub-pixel land-cover mapping; following SIH 26142 we apply it to enhancing the image itself.